

Experiment 8: Character Arrays and Searching

Objective: To learn how to manipulate character arrays and search for keys in arrays.

Questions:

1. Write a C program to find the length of a string using a loop (without using strlen()).

Sample Input:

Enter a string: Computer

Sample Output:

Length of string = 8

2. Write a C program to count vowels and consonants in a string.

Sample Input:

Enter a string: Education

Sample Output:

Vowels = 5

Consonants = 4

3. Write a C program to search for a character in a string and display its position.

Sample Input:

Enter a string: Programming

Enter character to search: g

Sample Output:

Character found at position 4

Character found at position 11

4. Write a C program to remove duplicate characters from a string without using any library string functions.

Sample Input:

Enter string: programming

Sample Output:

After removing duplicates: progamin

5. Write a C program to find the longest word in a sentence.

Sample Input:

Enter sentence: Data structure and algorithms

Sample Output:

Longest word: algorithms

Length: 10

6. Write a C program to remove multiple spaces between words and keep only single space.

Sample Input:

Enter string: C programming is fun

Sample Output:

C programming is fun

7. Write a C program to search for a given number in an integer array using linear search.

Sample Input:

Enter number of elements: 5

Enter elements: 10 20 30 40 50

Enter key to search: 30

Sample Output:

Element found at position 3

8. Write a C program to perform binary search on a sorted array using recursion.**Sample Input:**

Enter number of elements: 5

Enter sorted elements: 10 20 30 40 50

Enter key to search: 40

Sample Output:

Element found at position 4